AMAN PARGANIHA

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Summary

AI/ML Engineer (M.Tech, AI/ML) specializing in applied LLM systems, Retrieval-Augmented Generation (RAG), and multimodal/document AI, with hands-on experience in large-scale dataset curation, model benchmarking, and event-driven, containerized ML pipelines.

Education

International Institute of Information Technology Naya Raipur — India
Master of Technology in Computer Science & Engineering (AI/ML) | CGPA: 8.0/10.0 — *Aug 2025 – Aug 2027*

- Coursework: Generative AI, Deep Learning, Computer Vision, Data Science & Visualization, Cloud Computing

Shri Shankaracharya Technical Campus Bhilai — India
Bachelor of Technology in Computer Science & Engineering | CGPA: 8.22/10.0 — *Aug 2019 – Aug 2023*

- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Technical Skills

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformer architectures, Multimodal models, Fine-tuning, Model Evaluation
LLM & Document AI: LlamaIndex, LangChain, OpenAI API (GPT-4o / 4o-mini), RAG pipelines, vector databases (Qdrant), event-driven orchestration (Inngest), embedding strategies, OCR
MLOps & Deployment: Docker, AWS(EC2), FastAPI, REST APIs, end-to-end ML pipelines, reproducible experimentation, Git
Data Engineering: Pandas, NumPy, ETL pipelines, dataset curation, feature engineering, benchmarking, SQL, NoSQL
Programming: C++ (primary), Python, SQL — clean, modular, production-quality code
Research & Engineering: Paper reading & implementation, NLP, Feature Engineering, Debugging, Unit Testing, Code Reviews

Projects

Event-Driven RAG Agent for Document Ingestion & Q&A | Code | *Python, FastAPI, Inngest, LlamaIndex, Qdrant, OpenAI*

- Built an event-driven RAG system that asynchronously ingests and parses multi-page PDF documents and answers natural-language queries with grounded, citation-backed responses.
- Architected durable, fault-tolerant ingestion with Inngest (independently retryable parse → embed → store steps) over a semantic-retrieval layer using LlamaIndex chunking, OpenAI `text-embedding-3-large` embeddings, and a Qdrant vector store.
- Integrated OpenAI gpt-4o-mini with context-aware prompting for grounded answers over indexed documents; designed the pipeline to be modular and easily reconfigurable across chunking and embedding strategies.
- Exposed the workflow through a FastAPI backend with a Streamlit chat UI (Qdrant run via Docker) — demonstrating a decoupled, observable, end-to-end pipeline.

Multimodal Credit Risk Analysis using SEC XBRL & NLP | Code | *Python, Pandas, NumPy, Scikit-learn, NLTK/VADER, Git*

- Engineered an end-to-end pipeline that parses ∼41M raw SEC XBRL records (2022–2024, 12 quarters) into a curated multimodal dataset of ∼35,000 companies with 47 features (34 financial + 13 NLP-derived).
- Trained and benchmarked four ML models (Random Forest, Gradient Boosting, Logistic Regression, SVM) for binary and multi-class corporate credit-rating prediction, reaching 97.9% (binary) and 93.8% (multi-class) accuracy with a Random Forest on financial features.
- Engineered 13 NLP features (sentiment, risk, safety, uncertainty, readability) from filing text using NLTK/VADER; built modular, reproducible pipelines with logged model-comparison reports and a per-feature-set evaluation (financial-only vs. combined) to validate results.

Open Source Contributions

Matplotlib | PR #31422 merged into main (v3.11.0) | Reviewed by core maintainer *timhoffm*

- Resolved issue #26620 by improving legend parameter documentation; PR reviewed and approved by core maintainer timhoffm and merged into the v3.11.0 release — navigating a large, widely-used open-source codebase to deliver a production-quality contribution.

JupyterLab | PR #18668 approved | PR #18610 merged

- Improved user-facing documentation and added Playwright end-to-end test coverage; both PRs cleared multi-round review with core maintainers, reflecting attention to testing rigor and reproducibility.

Certifications & Achievements

Google × Kaggle — 5-Day AI Agents Intensive: Hands-on training in agentic LLM workflows, LlamaIndex/LangChain orchestration, and RAG system design.
GATE CSE: Ranked Top 7.8% nationwide (out of ∼2.1 lakh candidates) — secured M.Tech (AI/ML) at IIIT Naya Raipur.
Problem Solving: 200+ LeetCode problems solved; consistently top-ranked in programming coursework.